MICROPROCESSOR AND EMBEDDED SYSTEMS

UNIT 1 -QUESTION BANK

1.	With a Block diagram explain the architecture of **8086 microprocessor**. (10 Marks)
2.	Indicate the **addressing modes** of the **destination operand** and calculate its **physical address** for the following instructions of **Intel 8086 microprocessor**: i) ADD [DI], AL ii) SUB BLOCK, AX iii) AND [BX+DI], CL iv) ADC [BP+100H], AX Assume **CS = 1000H, DS = 2000H, ES = 3000H, SS = 4000H, BX = 3FFFH, DI = 5A00H, SP = 64A0H, IP = A000H, BP = 69B0H, BLOCK = 4000H**. (10 Marks)
3.	Write an **assembly level program** to **search the element** in a given array of bytes by applying the **linear search algorithm**. (8 Marks)
4.	Illustrate the following **Assembler directives** with proper examples: i) ASSUME ii) OFFSET iii) EQU iv) MACRO (4 Marks)
5.	Explain the following **instructions** with appropriate examples: i) MOV ii) PUSH iii) IN iv) XCHG V) IMUL VI) POP(8 Marks)
6.	Illustrate the **flag register of 8086 microprocessor** with a neat diagram. (6 Marks)
7.	Write an **Assembly level program** to obtain the **average of two numbers**. (6 Marks)
8.	Describe the **logical shift and rotate instructions** with syntax.
9.	Explain the concept of segmented memory and different types of segment registers available in 8086 Microprocessors. Show with an example how physical address is generated from logical address. (8 Marks)
10.	What is an addressing mode? List out any 6 addressing modes and explain with examples. (8 Marks)

11.	With examples, explain shift and rotate instructions available in 8086. (8 Marks)
12.	Explain with example working of the instructions: i) AAM ii) DAA (8 Marks)
13.	Write an ALP to read a character from the keyboard and display it with proper messages. (4 Marks)
14.	Explain the different **general-purpose and flag registers** of the **8086 microprocessor** with a neat diagram. (8 Marks)
15.	Explain the different **memory addressing modes** of the **8086 microprocessor** with an example. (6 Marks)
16.	Explain the following **8086 instructions** with an example for each: i) **DAA** ii) **AAA** (6 Marks)
17.	At a certain instant during the execution of a program, the **8086 processor** has the following data in the registers: **AX = 1234H, BX = 5678H, SI = ABCDH, DI = CDEFH, CS = 3456H, IP = 789AH, DS = ES = 4567H** Identify the **addressing modes** used and work out the **addresses of source and destination** of the data when each of the following instructions is executed: i) **MOV [BX], AX** ii) **MOV [BX+DI+3456H], 9ABCH** iii) **MOV AX, [9ABC]** iv) **MOV CX, [SI+BX]** (8 Marks)
18.	Explain the **advantages of segmentation** in **8086**. Illustrate with an example how the **physical address** is generated for the **code segment**. (6 Marks)
19.	Explain the **working of the Bus Interface Unit (BIU)** of the **8086 architecture** with a diagram. (6 Marks)
20.	Explain the following **assembler directives**: i) **DD** ii) **EVEN** iii) **INCLUDE** iv) **OFFSET** V) DB
21.	Develop an **ALP (Assembly Language Program)** to **search a key element** in a **list of 'n' 8-bit numbers** using the **Linear Search algorithm**. Display the **position** of the

	element if found; otherwise, display that the element is not found. (8 Marks)
22.	How do various **Shift Instructions** work? Differentiate between **arithmetic** and **logical shift instructions**. Explain with examples. (8 Marks)
23.	Develop an **ALP with procedures** to perform the following: i) Read a **string** from the **keyboard**, terminated by **carriage return** ii) Display the **string** iii) Display the **length** of the string (8 Marks)
24.	Explain the **flag register** along with the **register format** with respect to **Intel 8086**. (8 Marks)
25.	Develop an **ALP (Assembly Language Program)** to **reverse a given string** and check whether it is a **palindrome or not**. (8 Marks)
26.	List and explain the **various addressing modes** available in **Intel 8086**. (8 Marks)
27.	Explain the following **8086 Assembler Directives** with an example for each: i) **ASSUME** ii) **SEGMENT** iii) **INCLUDE** iv) **DB** (8 Marks)
28.	Explain the role of the following **4 directives**: **ASSUME, END, PROC, ENDS** (4 Marks)
29.	Write an ALP to read a character from keyboard and display it with proper message. If the entered character is ‘I’ , then print a message saying “INTEL 80886 PROCESSOR”.
30.	Check whether the following 8086 instructions are valid or not. Explain the reason if they are valid or not. i) ADD AX, 100H ii) MOV BL, AX iii) ADD CX, 1234H iv) MOV DS,5000H v) OUT 9000H,AX vi) ADD [5000H],[3000H]